2 System

Objectives

Upon completion of this chapter, the user will be able to understand:

1. Types of General system
2. Types of LTI system
3. Calculate output of LTI system using convolution
4. Properties of the convolution operator
5. Obtain convolution of two signals using various methods
6. LTI systems in cascade and parallel
7. Test your understanding questions
8. Note the common mistakes usually committed.

Introduction

In the previous chapter, we developed a basic understanding of signals using elementary signals and its properties. In this chapter, we will learn about different types of systems and how they are classified based on their dependency on the dependent and independent variables. In addition to that, how the output of a LTI system is obtained with the help of time-domain operation, i.e., convolution operator and its properties exclusively.

LTI and general system:

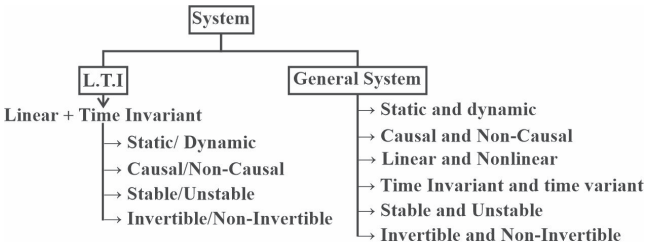
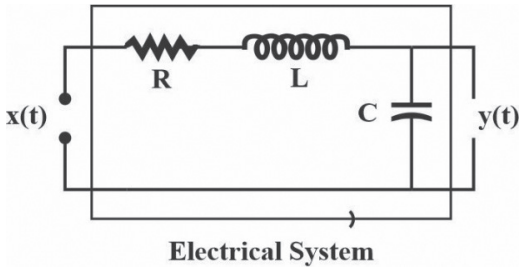


Fig. 2.1

System:

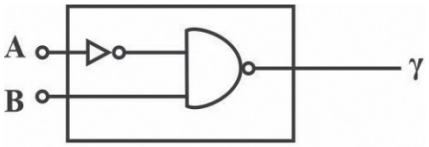
Is meaningful interconnection of components (R, L, and C) and devices (Diodes, BJTs, logic gates) which process input signal and generate an output signal

1.



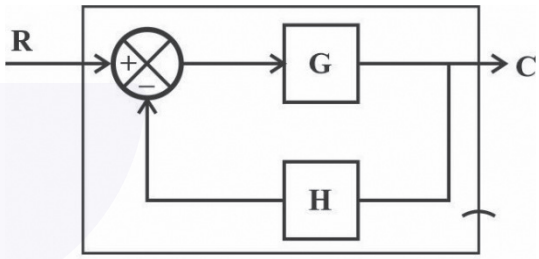
Electrical System

2.



Digital System

3.



Control System

Fig. 2.2

Block level representation of a system:

1.

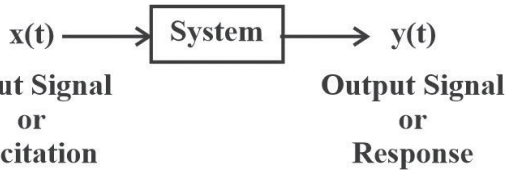


Fig. 2.3

2. Above system (S/S) is SISO system (single input-single output)

Mathematical representation of a system:

1.

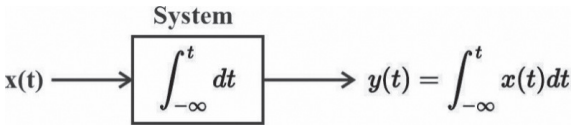


Fig. 2.4

2. Equation relating input and output defines a system mathematically.

$$y(t) = \int_{-\infty}^{t} x(t)dt$$

Important observation:

1.

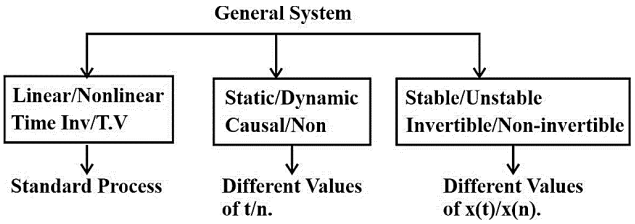


Fig. 2.5

2. Time relation:

   $y(2) = x(4) \rightarrow$ Output depends on the future value of input.

   $y(0) = x(0) \rightarrow$ Output depends on present value of input.

   $y(4) = x(2) \rightarrow$ Output depends on past value of input.

General System

Static and dynamic S/S:

Static S/S: "Present output must depend on present value of input only."

Dynamic S/S: Which is not a static system is a dynamic system.

Example 2.1: Identify the static and dynamic nature of the following systems.

a) $y(t) = 10x(t)$

Solution:

$t = 2 \rightarrow y(2) = 10x(2)$: Output depends on present value of input.

$t = 0 \rightarrow y(0) = 10x(0)$: Output depends on present value of input.

$t = -2 \rightarrow y(-2) = 10x(-2)$: Output depends on present value of input.

$t = \frac{1}{3} \rightarrow y\left(\frac{1}{3}\right) = 10x\left(\frac{1}{3}\right)$: Output depends on present value of input.

$t = \frac{-1}{3} \rightarrow y\left(\frac{-1}{3}\right) = 10x\left(\frac{-1}{3}\right)$: Output depends on present value of input.

Thus $y(t) = 10x(t)$ is static system

Note: Any operation (shifting, scaling, reversal) on the independent variable of x(t), which is time, leads to a dynamic system.

Examples of static system:

1. $\mathbf{y(t) = f[x(t)]}$
2. $\mathbf{y(t) = \ln\{x(t)\}}$

Examples of dynamic systems:

1. $y(t) = x(-t)$
2. $y(t) = x(t-2)$

Example 2.2: Identify the static or dynamic nature of the system $y(t) = x(t) + u(t+2)$

$u(\cdot) \rightarrow$ *unit* step, and x (.) is input

Solution:

$t = 0 : y(0) = x(0) + u(2)$: Output depends on present value of input.

$t = 2 \quad y(2) = x(2) + u(4)$: Output depends on present value of input.

Note: relation between y(t) and x(t) must be noticed. One should not give a conclusion based on the relation between y(t) and u(t), because u(t) is not input.

Special category:

1. $y(t) = \text{Even}\{x(t)\} = \frac{x(t) + x(-t)}{2}$

   $y(n) = \text{Even}\{x(n)\} = \frac{x(n) + x(-n)}{2}$

   Dynamic System

2. $y(t) = \text{odd}\{x(t)\} = \frac{x(t) - x(-t)}{2}$

   $y(n) = \text{odd}\{x(n)\} = \frac{x(n) - x(-n)}{2}$

   Dynamic System

3. $y(t) = \text{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2}$

   $y(n) = \text{Re}\{x(n)\} = \frac{x(n) + x^*(n)}{2}$

   Static System

4. $y(t) = \text{Im}\{x(t)\} = \dfrac{x(t) - x^*(t)}{2j}$

$y(n) = \text{Im}\{x(n)\} = \dfrac{x(n) - x^*(n)}{2j}$

Static System

5. $y(t) = \text{Even Conjugate}\{x(t)\}$

$= \dfrac{x(t) + x^*(-t)}{2}$

$y(n) = \text{Even Conjugate}\{x(n)\}$

$= \dfrac{x(n) + x^*(-n)}{2}$

Dynamic System

6. $y(t) = \text{Odd Conjugate}\{x(t)\}$

$= \dfrac{x(t) - x^*(-t)}{2}$

$y(n) = \text{Odd Conjugate}\{x(n)\}$

$= \dfrac{x(n) - x^*(-n)}{2}$

Dynamic System

7. $y(t) = \dfrac{dx(t)}{dt} \rightarrow$ Dynamic system

$\dfrac{d}{dt} \rightarrow$ Symbol or Operator

$y(t) = \dfrac{dx(t)}{dt} = \lim\limits_{\Delta t \to 0} \dfrac{x(t + \Delta t) - x(t)}{\Delta t}$

$= \lim\limits_{\Delta t \to 0} \dfrac{x(t) - x(t - \Delta t)}{\Delta t}$

$y(t) = \dfrac{dx(t)}{dt} = \lim\limits_{\Delta t \to 0} \dfrac{x(t) - x(t - \Delta t)}{\Delta t}$

8. First difference or successive difference

$y(n) = x(n) - x(n-1) \rightarrow$ Dynamic System

9. $y(t) = \int_{-\infty}^{t} x(\tau) d\tau \rightarrow y(t) = \left[m(\tau)\right]_{-\infty}^{t}$

$y(t) = m(t) - m(-\infty)$

10. Accumulator:

$y(n) = \sum\limits_{k=-\infty}^{n} x(k) \rightarrow$ Dynamic

$y(n) = x(n) + x(n-1) + x(n-2) + \cdots + x(-\infty)$

11. Piecewise defined function:

a) $y(t) = \begin{cases} x(2t): & t \geq 0 \\ x\left(\dfrac{t}{2}\right): & t < 0 \end{cases} \rightarrow$

Dynamic system

$t = 2: \quad y(2) = x(4)$

$t = 0: \quad y(0) = x(0)$

$t = -2: \quad y(-2) = x(-1)$

$t = 1/2: \quad y\left(\dfrac{1}{2}\right) = x(1)$

$t = \dfrac{-1}{2}: \quad y\left(\dfrac{-1}{2}\right) = x\left(\dfrac{-1}{4}\right)$

b) $y(n) = \begin{cases} x(n) & : n < 0 \\ 0 & : n = 0 \\ x(n-1) & : n > 0 \end{cases}$

$\rightarrow$ Dynamic System

$n = 2 \quad y(2) = x(1)$

$n = -2 \quad y(-2) = x(-2)$

Note:

Static system:

1. Memory less system
2. Electrical static systems contain resistors only.

Dynamic system:

1. Systems with memory
2. Electrical dynamic S/S contains L, C along with resistors.

Causal-non-causal system:

Causal system: "Output depends on either past or present value of input or both but not on future value of input."

Note:

1. All static systems are causal systems → Converse is not necessarily true.
2. Causal systems are practical systems.

Non-causal system:

→ Which is not a causal system.

→ All non-causal systems are dynamic→ converse is not necessarily true

Anti-causal system: Output shall depend on the future value of input only.

Example 2.3: Determine the causal and non-causal nature of the following system.

a) $y(t) = x(-t^2)$:

Solution:

$t = 2: y(2) = x(-4) \rightarrow$ Output depends on past value of input.

$t = -2: y(-2) = x(-4) \rightarrow$ Output depends on past value of input.

$t = 0: y(0) = x(0) \rightarrow$ Output depends on present value of input.

$t = \frac{1}{2}: y\left(\frac{1}{2}\right) = x\left(\frac{-1}{4}\right) \rightarrow$ Output depends on past value of input.

$t = \frac{-1}{2}: y\left(\frac{-1}{2}\right) = x\left(\frac{-1}{4}\right) \rightarrow$ Output depends on the future value of input.

Thus, $y(t) = x(-t^2)$ is non-causal system

b) $y(n) = x(-n^2)$

Solution:

$n = 2: y(2) = x(-4) \rightarrow$ Output depends on past value of input.

$n = 0: y(0) = x(0) \rightarrow$ Output depends on present value of input.

$n = -2: y(-2) = x(-4) \rightarrow$ Output depends on past value of input.

So, $y(n) = x(-n^2)$ is causal system

Special category:

1. $y(t) = Ev\{x(t)\} = \frac{x(t) + x(-t)}{2}$

$y(n) = Ev\{x(n)\} = \frac{x(n) + x(-n)}{2}$

Non – Causal System

2. $y(t) = odd\{x(t)\} = \frac{x(t) - x(-t)}{2}$

$y(n) = odd\{x(n)\} = \frac{x(n) - x(-n)}{2}$

Non – Causal System

3. $y(t) = \text{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2}$

$y(n) = \text{Re}\{x(n)\} = \frac{x(n) + x^*(n)}{2}$

Static → *Causal* System

4. $y(t) = \text{Im}\{x(t)\} = \frac{x(t) - x^*(t)}{2j}$

$y(n) = \text{Im}\{x(n)\} = \frac{x(n) - x^*(n)}{2j}$

Static → *Causal* System

5. $y(t) = EC\{x(t)\} = \frac{x(t) + x^*(-t)}{2}$

$y(n) = EC\{x(n)\} = \frac{x(n) + x^*(-n)}{2}$

Noncausal System

6. $y(t) = OC\{x(t)\} = \frac{x(t) - x^*(-t)}{2}$

$y(n) = OC\{x(n)\} = \frac{x(n) - x^*(-n)}{2}$

Noncausal System

7. $y(t) = \frac{dx(t)}{dt}$

$y(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}$

$= \lim_{\Delta t \to 0} \frac{x(t) - x(t - \Delta t)}{\Delta t.}$

$y(t) = \lim_{\Delta t \to 0} \frac{x(t) - x(t - \Delta t)}{\Delta t} = \frac{dx(t)}{dt} \rightarrow$ Causal system

8. $y(n) = x(n) - x(n-1) \rightarrow$ Causal system

9. a) Standard integrator:

$y(t) = \int_{-\infty}^{t} x(\tau) d\tau$

$y(t) = \left[m(\tau)\right]_{-\infty}^{t}$

$$y(t) = m(t) - m(-\infty)$$

So, standard integrators are causal systems.

b) Non-standard integrators:

$$y(t) = \int_{-\infty}^{-t^2} x(\tau)d\tau$$

$$y(t) = m(-t^2) - m(-\infty) \rightarrow \text{Non-causal system}$$

10. Standard accumulator

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

$$y(n) = x(n) + x(n-1) + x(n-2) + ...x(-\infty)$$

So, standard accumulators are causal systems.

Non-standard accumulator

(i) $$y(n) = \sum_{k=-\infty}^{|n|} x(k)$$

$$= x(|n|) + x(|n|-1) + x(|n|-2) + ...$$

Above system is a non-causal system.

11. Piecewise defined function:

$$y(t) = \begin{cases} x(2t): & t < 0 \\ x\left(\frac{t}{2}\right): & t \geq 0 \end{cases}$$

$t = +2 : \ y(2) = x(1)$

$\rightarrow$ Output depends on past value of input.

$t = 0 \ : \ y(0) = x(-1)$

$\rightarrow$ Output depends on past value of input.

$t = -2 : \ y(-2) = x(-4)$

$\rightarrow$ Output depends on past value of input.

$$t = \frac{1}{2} \ : \ y\left(\frac{1}{2}\right) = x\left(\frac{1}{4}\right)$$

$\rightarrow$ Output depends on past value of input.

$$t = \frac{-1}{2} \ : \ y\left(\frac{-1}{2}\right) = x\left(-\frac{1}{4}\right)$$

$\rightarrow$ Output depends on future value of input.

Above system is non-causal system.

Linear and nonlinear system:

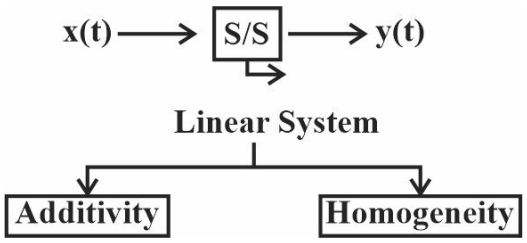


Fig. 2.6

Step 1: Identify the definition of system

$$x(t) \xrightarrow{S} y(t)$$

Step 2: $x_1(t) \xrightarrow{S} y_1(t)$, $x_2(t) \xrightarrow{S} y_2(t)$ $\oplus$ ...(i)

$$x_1(t) + x_2(t) \xrightarrow{S} y_3(t) \quad ...(ii)$$

(i) = (ii) $\rightarrow$ Additivity holds

Or

Additivity:

$f(A+B) = f(A) + f(B) \rightarrow$ f () is additive operator.

$f(A+B) \neq f(A) + f(B) \rightarrow$ f () is non additive operator.

S.3$\rightarrow$ Homogeneity:

Processing of input $Ax(t) \xrightarrow{S} y_4(t)$ ...(iii)

Mathematical calculation of $Ay(t)$ ...(iv)

If (iii) = (iv) for all possible values of A real or complex, then system is homogeneous.

$\rightarrow$ "For a system to be linear system it has to be both additive as well as homogeneous."

$\rightarrow$ Additivity + Homogeneity $\rightarrow$ Linearity

Example 2.4: Comment on the linearity of the system $y(t) = x(t) + 5$

Solution:

Step 1: $x(t) \xrightarrow{S} x(t) + 5$

Step 2:

$x_1(t) \xrightarrow{S} x_1(t) + 5$

$x_2(t) \xrightarrow{S} x_2(t) + 5$ $\oplus$ $x_1(t) + x_2(t) + 10$ ...(i)

$[x_1(t)+x_2(t)] \xrightarrow{S} [x_1(t)+x_2(t)]+5$ ....(ii)

$(i) \neq (ii)$ Non additive system → Nonlinear system

Step 3→

Processing of input:

$x(t) \xrightarrow{S} x(t)+5$

$[Ax(t)] \xrightarrow{S} [Ax(t)]+5$ ...(iii)

Mathematical calculation →

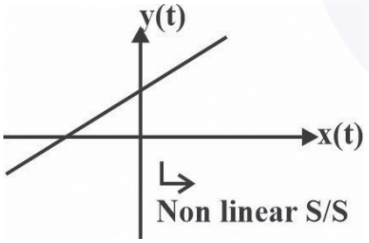
$y(t) = x(t)+5$

$Ay(t) = Ax(t)+5A$ ...(iv)

$(iii) \neq (iv) \rightarrow$ Non-homogeneous S/S→ Nonlinear system

Note:

1. Linear relation between y(t) and x(t) does not always lead to linear S/S.
2. $y(t) = mx(t)+c$



3. $y(t) = mx(t)$

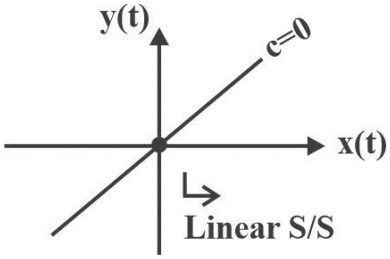


Fig. 2.7

Note:

y(t) = x[f(t)]
y(n) = x(f(n)) → Linear S/S

Examples of linear system:

$y(t) = x(-t)$

$y(t) = x(t^2)$

$y(t) = x(\ln t)$

$y(n) = x(|n|)$

5. Comment on the linearity of the system

$y(t) = |t|\,x(t)$

Step 1→ $x(t) \xrightarrow{S} |t|\,x(t)$

Step 2→

$x_1(t) \xrightarrow{S} |t|\,x_1(t)$

$\oplus\ |t|(x_1(t)+x_2(t))$ ..(i)

$x_2(t) \xrightarrow{S} |t|\,x_2(t)$

$[x_1(t)+x_2(t)] \xrightarrow{S} |t|[x_1(t)+x_2(t)]$ ...(ii)

(i) = (ii) → Additive system

Step 3→

Processing of input →

$x(t) \xrightarrow{S} |t|\,x(t)$

$[Ax(t)] \xrightarrow{S} |t|[Ax(t)]$ ...(iii)

Mathematical calculation →

$y(t) = |t|\,x(t)$

$Ay(t) = A|t|\,x(t)$ ...(iv)

(iii) = (iv) → Homogenous

Since, above system is both additive and homogeneous, hence system is a linear system.

Note:

y(t) = g(t)x(t) → Linear S/S

y(t) = g(t)x(h(t)) → Linear S/S

Examples of linear system:

y(t) = $|t^2 - 2t + 1|\,x(|\sin t|)$

y(t) = sint × $x(e^t)$

Chapter Summary

1. A linear equation between x(t) and y(t) may or may not represent linear S/S.

$y(t) = mx(t)+c$ → $c = 0 \rightarrow$ *linear* System; $c \neq 0 \rightarrow$ *Non*linear System

2. $y(t) = mx^k(t)+c$ *where* $k \neq 1$ always represents Nonlinear System

   $y(t) = x(f(t))$

3. $y(t) = g(t)x(t)$ → Linear system

   $y(t) = g(t)x(f(t))$

4. A non-linear operator between x(t) and y(t) makes the S/S non-linear.

(i) Polynominal Based	(ii) Expansion Based	(iii) Special Category
$y(t) = (x(t))^k$ $k \neq 1$	$y(t) = e^{x(t)}$ $y(t) = \ln\{x(t)\}$ $y(t) = \sin\{x(t)\}$ $y(t) = \cos\{x(t)\}$	$y(t) = \lvert x(t) \rvert$ $y(t) = x^*(t)$ $y(t) = \text{sgn}(x(t))$ $y(t) = u(x(t))$ $y(t) = \sqrt{x(t)}$

Special categories:

1. $y(t) = Ev\{x(t)\} = \dfrac{x(t) + x(-t)}{2}$

   $y(n) = Ev\{x(n)\} = \dfrac{x(n) + x(-n)}{2}$

   Linear System

2. $y(t) = odd\{x(t)\} = \dfrac{x(t) - x(-t)}{2}$

   $y(n) = odd\{x(n)\} = \dfrac{x(n) - x(-n)}{2}$

   Linear System

3. $y(t) = \text{Re}\{x(t)\} = \dfrac{x(t) + x^*(t)}{2}$

   $y(n) = \text{Re}\{x(n)\} = \dfrac{x(n) + x^*(n)}{2}$

   Non Linear System

4. $y(t) = \text{Im}\{x(t)\} = \dfrac{x(t) - x^*(t)}{2j}$

   $y(n) = \text{Im}\{x(n)\} = \dfrac{x(n) - x^*(n)}{2j}$

   Non Linear System

5. $y(t) = EC\{x(t)\} = \dfrac{x(t) + x^*(-t)}{2}$

   $y(n) = EC\{x(n)\} = \dfrac{x(n) + x^*(-n)}{2}$

   Non Linear System

6. $y(t) = OC\{x(t)\} = \dfrac{x(t) - x^*(-t)}{2}$

   $y(n) = OC\{x(n)\} = \dfrac{x(n) - x^*(-n)}{2}$

   Non Linear System

7. $y(t) = \dfrac{dx(t)}{dt} \rightarrow$ Linear system

8. $y(n) = x(n) - x(n-1) \rightarrow$ Linear system

9. a) Standard integrator:

   $y(t) = \int_{-\infty}^{t} x(\tau)d\tau \rightarrow$ Linear system.

10. Standard accumulator

    $y(n) = \sum_{k=-\infty}^{n} x(k) \rightarrow$ Linear system

11. Piecewise defined systems: Always check by basic method

12. OPAMP as a comparator:

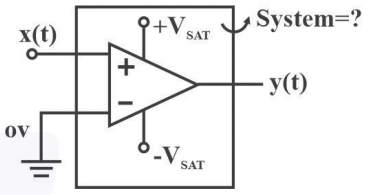


Fig. 2.8

$$y(t) = \begin{cases} +V_{SAT} & : x(t) > 0 \\ -V_{SAT} & x(t) < 0 \end{cases}$$

$y(t) = +V_{SAT}\,\text{sgn}(x(t)) \rightarrow$

Non-linear system

Time invariant, time variant system:

1. Shift invariant system
2. Constant parameter system

Procedure:

Step 1 → Delayed input processing:

$x(t) \xrightarrow{S} y(t)$

$x_1(t) \xrightarrow{S} y_1(t)$

For $x_1(t) = x(t - t_0)$ calculate $y_1(t)$ ...(i)

Step 2 → Mathematical calculation of

$y(t - t_0)$ ... (ii)

If $(i) = (ii) \rightarrow$ Then system will be time invariant system

Example 2.4: Comment on the time invariant nature of the system $y(t) = x(2t)$.

Solution:

Step 1→

$x(t) \xrightarrow{S} x(2t)$

$x_1(t) \xrightarrow{S} x_1(2t)$

Let $x_1(t) = x(t - t_0)$

So, $x_1(2t) = x(2t - t_0)$ ... (i)

Step 2→

$y(t) = x(2t)$

$y(t - t_0) = x(2(t - t_0) = x(2t - 2t_0)$ ...(ii)

$(i) \neq (ii) \rightarrow$ Above system is time variant

7. Comment on the time invariant nature of the system $y(t) = x(-t)$

Solution:

Step 1→

$x(t) \xrightarrow{S} x(-t)$

$x_1(t) \xrightarrow{S} x_1(-t)$

$Let\ x_1(t) = x(t - t_0)$

$so\ x_1(-t) = x(-t - t_0)$ ...(i)

Step 2→

$y(t) = x(-t)$

$y(t - t_0) = x(-(t - t_0)) = x(-t + t_0)$ ...(ii)

$(i) \neq (ii) \rightarrow$ Above system is time variant

Important observations:

1. $y(t) = x(f + (t))$
   → Linear
   → *T.V.*
2. $y(t) = g(t)x(t)$
   → Linear
   → *T.V.*
3. $y(t) = x(t) + f(t)$
   → *Non*linear
   → *T.V.*
4. $y(t) = f(x(t))$
   → *Non*linear if f is Nonlinear operator
   → *T.V.*

Special categories:

1. $y(t) = Ev\{x(t)\} = \frac{x(t) + x(-t)}{2}$

   $y(n) = Ev\{x(n)\} = \frac{x(n) + x(-n)}{2}$

   Time Variant

2. $y(t) = odd\{x(t)\} = \frac{x(t) - x(-t)}{2}$

   $y(n) = odd\{x(n)\} = \frac{x(n) - x(-n)}{2}$

   Time variant

3. $y(t) = \text{Re}\{x(t)\} = \frac{x(t) + x^*(t)}{2}$

   $y(n) = \text{Re}\{x(n)\} = \frac{x(n) + x^*(n)}{2}$

   Time Invariant

4. $y(t) = \text{Im}\{x(t)\} = \frac{x(t) - x^*(t)}{2j}$

   $y(n) = \text{Im}\{x(n)\} = \frac{x(n) - x^*(n)}{2j}$

   Time Invariant

5. $y(t) = EC\{x(t)\} = \frac{x(t) + x^*(-t)}{2}$

   $y(n) = EC\{x(n)\} = \frac{x(n) + x^*(-n)}{2}$

   Time variant

6. $y(t) = OC\{x(t)\} = \frac{x(t) - x^*(-t)}{2}$

   $y(n) = OC\{x(n)\} = \frac{x(n) - x^*(-n)}{2}$

   Time variant

7. $y(t) = \frac{dx(t)}{dt} = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} \rightarrow$

   Time Invariant

8. $y(n) = x(n) - x(n - 1) \rightarrow$ Time Invariant
9. Standard integrator

   a) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

Method 1: $y(t) = m(t) - m(-\infty)$

Step 1→

$m(t) \to m(t) - m(-\infty)$

$m_1(t) \to m_1(t) - m_1(-\infty)$

$Let\ m_1(t) = m(t - t_0)$

$so\ m_2(t) - m_1(-\infty)$

$= m(t - t_0) - m(-\infty - t_0) = m(t - t_0)$

$-m(-\infty).....(i)$

Step 2→

$y(t - t_0) = m(t - t_0) - m(-\infty)$... (ii)

$(i) = (ii) \to$ Above system is time invariant

Method 2: $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

Step 1→

$x(t) \xrightarrow{S} \int_{-\infty}^{t} x(\tau)d\tau$

$x_1(t) \xrightarrow{S} \int_{-\infty}^{t} x_1(\tau)d\tau$

$Let\ x_1(t) = x(t - t_0)$

$so\quad x_1(\tau) = x(\tau - t_0)$

$\int_{-\infty}^{t} x_1(\tau)d\tau = \int_{-\infty}^{t} x(\tau - t_0)d\tau.....(i)$

Step 2→

$y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

$y(t - t_0) = \int_{-\infty}^{t-t_0} x(\tau)d\tau$ ...(ii)

i) $\int_{-\infty}^{t} x(\tau - t_0)d\tau = m(t - t_0) - m(-\infty)$

ii) $\int_{-\infty}^{t-t_0} x(\tau)d\tau = m(t - t_0) - m(-\infty)$

Via substitution:

a) $\int_{-\infty}^{t} x(\tau - t_0)d\tau$

$\tau - t_0 = p$

$d\tau = dp$

$(i)\int_{-\infty}^{t-t_0} x(\tau)d\tau$

$(ii)\int_{-\infty}^{t-t_0} x(\tau)d\tau$

$(i) = (ii) \to$ Above system is time invariant

10. Non-standard integrator:

$\to y(t) = \int_{-\infty}^{2t} x(\tau)d\tau$

$y(t) = m(2t) - m(-\infty) \to$ Time variant system

$\to y(t) = \int_{-\infty}^{|t|} x(\varepsilon)d\tau$

$y(t) = m(|t|) - m(-\infty)$

$\to$ Time Variant system

11. Standard accumulator:

$y(n) = \sum_{k=-\infty}^{n} x(k)$

Step 1→

$x(n) \xrightarrow{S} \sum_{k=-\infty}^{n} x(k) \quad x_1(n) \xrightarrow{S} \sum_{k=-\infty}^{n} x_1(k)$

$Let\ x_1(n) = x(n - n_0)$

$so\quad x_1(k) = x(k - n_0)$

$\sum_{k=-\infty}^{n} x_1(k) = \sum_{k=-\infty}^{n} x(k - n_0).....(i)$

Step 2→

$y(n) = \sum_{k=-\infty}^{n} x(k)$

$y(n - n_0) = \sum_{k=-\infty}^{n-n_0} x(k).....(ii)$

$k - n_0 = p$

$(i)\sum_{p=-\infty}^{n-n_0} x(p) = \sum_{p=-\infty}^{n-n_0} x(k)$

$(ii)\sum_{p=-\infty}^{n-n_0} x(k)$

$(i) = (ii) \to$ Above system is time invariant

12. Non-standard accumulator

$\to y(n) = \sum_{k=-\infty}^{2n} x(k)$

$y(n) = x(2n) + x(2n-1) + \cdots x(-\infty)$

$\to$ Time Variant

$y(n) = \sum_{K=-\infty}^{n^2} x(k)$

$y(n) = x(n^2) + x(n^2 - 1) + \cdots$

$\to$ Time variant

13. Piecewise defined system:

$$y(n)=\begin{cases} x(n-1) & n<0 \\ 0 & n=0 \\ x(n) & n>0 \end{cases}$$

Step 1→

$$x(n)\xrightarrow{S}\begin{cases} x(n-1) & n<0 \\ 0 & n=0 \\ x(n) & n>0 \end{cases}$$

$$x_1(n)\xrightarrow{S}\begin{cases} x_1(n-1) & n<0 \\ 0 & n=0 \\ x_1(n) & n>0 \end{cases}$$

Let $x_1(n)=x(n-n_0)$

$$so \quad \begin{cases} x(n-n_0-1): & n<0 \\ 0: & n=0 \\ x(n-n_0): & n>0 \end{cases} \ldots\ldots\ldots\ldots\ldots\ldots\ldots(i)$$

Step 2

$$y(n)=\begin{cases} x(n-1) & n<0 \\ 0 & n=0 \\ x(n) & n>0 \end{cases}$$

$$y(n-n_0)=\begin{cases} x(n-n_0-1) & : n-n_0<0 \\ 0 & : n-n_0=0 \\ x(n-n_0) & : n-n_0>0 \end{cases} \ldots(ii)$$

$(i)\neq(ii)\rightarrow$ Above system is time variant

Note: All piecewise defined functions are always T.V

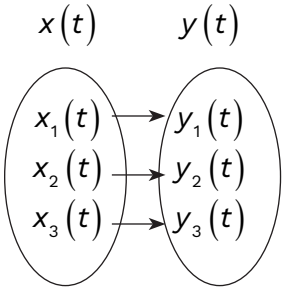
Invertible and non-invertible systems:

Method 1: Graphical method

Method 2: Tabular method

Invertible system: There must be a one-one mapping between x(t) and y(t).

System must result in different outputs for each and every possible different input.



Method 1: Graphical method

S.1→ Plot y(t) vs x(t)

S.2→ Draw a horizontal line

S.3→ If the intersection of y(t) vs x(t) and the horizontal line is

1. At signal point → Invertible S/S.
2. At multiple points → Non-invertible S/S.

Note: For a system to be invertible, the graph of y(t) vs x(t) must be strictly monotonic in nature (either strictly ↑ or strictly ↓)

7. Comment on the invertibility of the system $y(t)=10x(t)$

Solution:

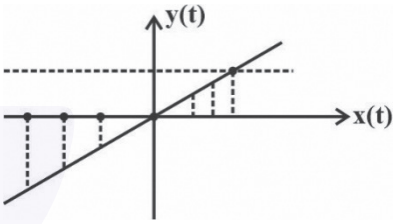


Fig. 2.9

Since, the graph of y(t) vs x(t) is strictly monotonic in nature, so the system is invertible.

Note:

→ If a system is invertible, then its inverse system will exist.

→ Inverse system may not be invertible.

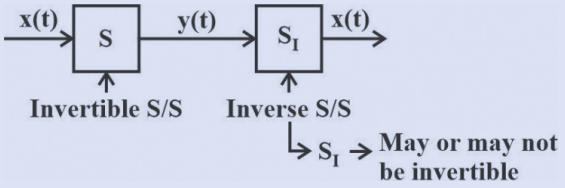


Fig. 2.10

Procedure to calculate inverse system:

Step 1→ Replace $x \leftrightarrow y$

Step 2→ Write y completely in terms of x

7. $y(t)=\int_{-\infty}^{t} x(\tau)d\tau$.

Solution:

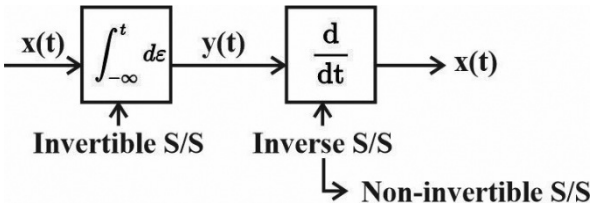


Fig. 2.11

a) Invertible S/S

b) Inverse S/S exist

$x(t) = \int_{-\infty}^{t} y(\tau) d\tau$

Inverse S/S will be: $y(t) = \frac{dx(t)}{dt}$

Important category:

8. Comment on the invertibility of the system $y(t) = \cos t \; x(t)$.

Solution:

$y(t) = \cos t \; x(t) \rightarrow$ Non-invertible

$x(t) = \delta\left(t - \frac{\pi}{2}\right)$:

$y(t) = \cos t \delta\left(t - \frac{\pi}{2}\right) = \cos\frac{\pi}{2}\delta\left(t - \frac{\pi}{2}\right) = 0$

$x(t) = \delta\left(t - \frac{3\pi}{2}\right)$: $\quad y(t) = \cos t \delta\left(t - \frac{3\pi}{2}\right) = 0$

Since, different inputs produce the same outputs, so the system is non-invertible.

Stable-unstable system:

Criteria: Bounded input-Bounded output criteria (BIBO).

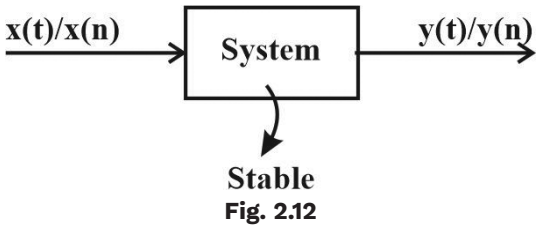


Fig. 2.12

For every bounded x(t)/x(n) , y(t)/y(n) must be bounded

Procedure:

Method 1: Inequality method

1. $|x_1(t)x_2(t)| = |x_1(t)||x_2(t)|$
2. $|x_1(t) + x_2(t)| \le |x_1(t)| + |x_2(t)|$
3. $\left|\int x_1(t)x_2(t)dt\right| \le \int |x_1(t)||x_2(t)|dt$
4. $\left|\sum x_1(n)x_2(n)\right| \le \sum |x_1(n)||x_2(n)|$

Step 1→ Let x(t) is bounded:

$|x(t)| \le M < \infty \quad \forall -\infty < t < \infty$

$|x(-at + b)| \le M < \infty \forall -\infty < t < \infty$

Step 2→ Calculate $|y(t)|$

Convert $|y(t)| \le N$ in this form.

S.3→ Test N for $\forall -\infty < t < \infty$

Note:

If N $\rightarrow \infty$: *system is* **unstable**

/f N $\rightarrow$ **finite** : *system is* **stable**

9. Comment on the stability of the system $y(t) = 10x(t)$.

Solution:

Step 1→ $|x(t)| \le M < \infty$

Step 2→ $|y(t)| = |10x(t)|$

$|y(t)| = 10|x(t)| \le 10M$

$|y(t)| \le 10M$

Step 3→ $|y(t)| \le 10M \rightarrow N = 10M$

$N = 10M$ for $-\infty < t < \infty$

$N = 10M$ is finite as $-\infty < t < \infty$

So above system is BIBO stable.

Note:

1. **y(t) =** $\int_{-\infty}^{t}$ **x(**τ**)sink**τ**d**τ

   x(t) = sinkt $\rightarrow$ **BI** $\rightarrow$ **y(t)** $\rightarrow$ **U***nbounded*

2. **y(t) =** $\int_{-\infty}^{t}$ **x(**τ**)cosk**τ**d**τ

   x(t) = coskt $\rightarrow$ **BI** $\rightarrow$ **y(t)** $\rightarrow$ **U***nbounded*

Linear time invariant system (LTI system):

Linear system:

1. $x_1(t) \xrightarrow{S} y_1(t)$

   $x_2(t) \xrightarrow{S} y_2(t)$

   $x_1(t) + x_2(t) \xrightarrow{S} y_1(t) + y_2(t)$

2. $Ax_1(t) \xrightarrow{S} Ay_1(t)$

   $Bx_2(t) \xrightarrow{S} By_2(t)$

   $Ax_1(t) + Bx_2(t) \xrightarrow{S} Ay_1(t) + By_2(t)$

Time invariant:

$x(t) \xrightarrow{S} y(t)$

$x(t-T) \xrightarrow{S} y(t-T)$

Linear Time Invariant (LTI):

$x(t) \longrightarrow y(t)$

$Ax(t) + Bx(t-T) \longrightarrow Ay(t) + By(t-T)$

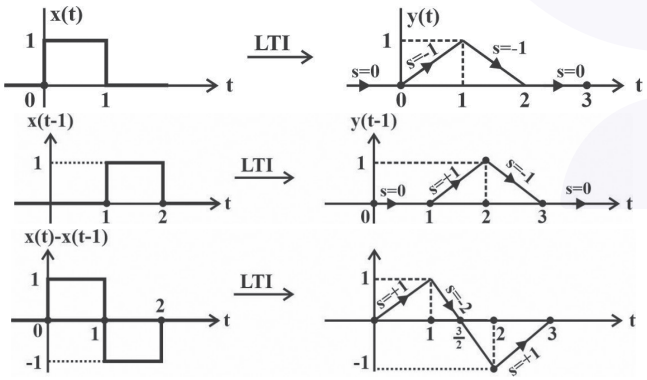


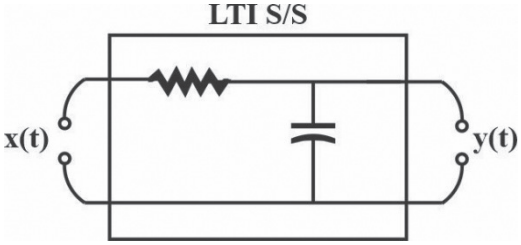
Fig. 2.13

Continuous time LTI system:



Fig. 2.14

Lab:



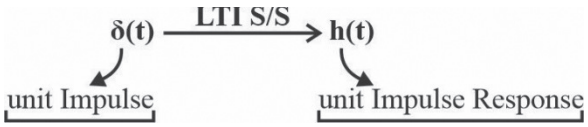


Fig. 2.15

$\delta(t) \xrightarrow{LTI\ S/S} h(t)$

$\delta(t-1) \xrightarrow{LTI\ S/S} h(t-1)$

$x(1)\delta(t-1) \xrightarrow{LTI\ S/S} x(1)h(t-1)$ ...(i)

$x(2)\delta(t-2) \xrightarrow{LTI\ S/S} x(2)h(t-2)$ ...(ii)

$x(3)\delta(t-3) \xrightarrow{LTI\ S/S} x(3)h(t-3)$ ...(iii)

$\sum_{\tau} x(\tau)\delta(t-\tau) \xrightarrow{LTI} \sum_{\tau} x(\tau)h(t-\tau)$

$\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \xrightarrow{LTI} \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$\int_{-\infty}^{\infty} x(t)\delta(t-\tau)d\tau \xrightarrow{LTI} \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$x(t)\int_{-\infty}^{\infty} \delta(t-\tau)d\tau \xrightarrow{LTI} \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$x(t) \xrightarrow{LTI} \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

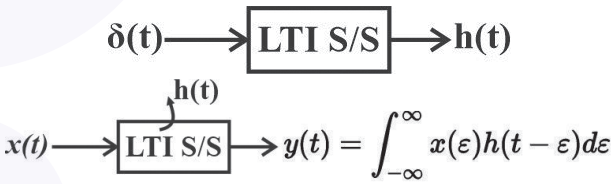


Fig. 2.16

h(t):

Unit impulse response of LTI system.

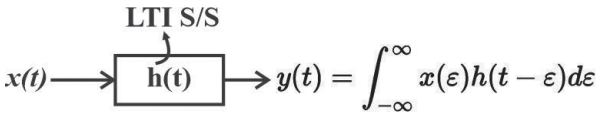
OR

Mathematical representation of LTI system

OR

LTI system parameter

Calculation of output of LTI S/S:



$$y(t) = x(t) * h(t)$$

Fig. 2.17

$*$ → Convolution operator

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$
$$= \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

Note: Convolution operator is only applicable for LTI system.

Properties of convolution operator:

1. $A * B = A \times B$
2. Commutative: $x(t) * h(t) = h(t) * x(t)$
3. Distributive: $x(t) * \left[h_1(t) + h_2(t)\right]$
   $= x(t) * h_1(t) + x(t) * h_2(t)$
4. Associative: $x(t) * \left(h_1(t) * h_2(t)\right)\Big)$
   $= \left[x(t) * h_1(t)\right] * h_2(t)$

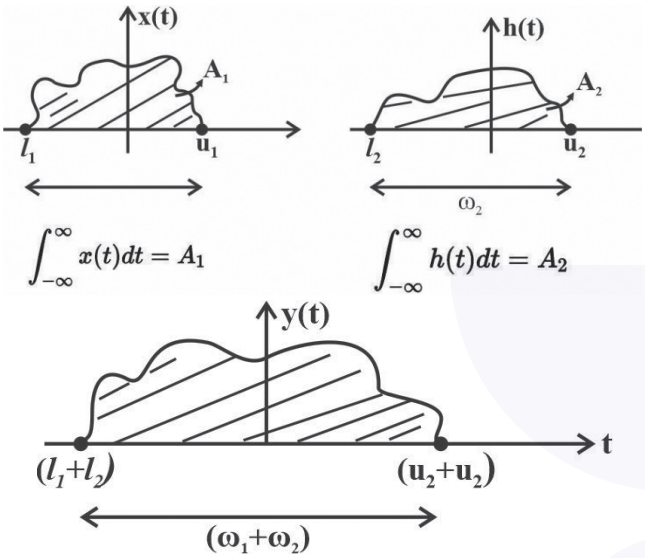


Fig. 2.17

5. $y(t) = x(t) * h(t)$

   $\int_{-\infty}^{\infty} y(t)dt = A_1 \times A_2$
6. $x(t) * h(t) = y(t)$
   $x(t-a) * h(t-b) = y(t-a-b)$
7. $x(t) * h(t) = y(t)$
   $x(-t) * h(-t) = y(-t)$

$x(t)$	$h(t)$	$y(t)$
E	E	E
E	0	0
0	E	0
0	0	E

8. $x(t) * h(t) = y(t)$
   $x(at) * h(at) = \frac{1}{|a|} y(at)$

   Scaling factor should be the same.
9. $x(t) * h(t) = y(t)$
   $Ax(t) * Bh(t) = AB(x(t) * h(t)) = ABy(t)$
10. $x(t) * h(t) = y(t)$

    $\frac{dx(t)}{dt} * h(t) = \frac{dy(t)}{dt}$

    $x(t) * \frac{dh(t)}{dt} = \frac{dy(t)}{dt}$

    $\frac{d^k(t)}{dt^k} * \frac{d^m h(t)}{dt^m} = \frac{d^{m+k} y(t)}{dt^{m+k}}$

Standard results:

1. $x(t) * \delta(t) = x(t)$
2. $x(t-a) * \delta(t-b) = x(t-a-b)$
3. $\delta(t) * \delta(t) = \delta(t)$
4. $\delta(t) * \delta(t) * \delta(t) * \cdots = \delta(t)$
5. $\delta(t-a) * \delta(t-b) = \delta(t-a-b)$
6. $x(t) * u(t) = \int_{-\infty}^{t} x(\tau)d\tau$
7. $\delta(t) * u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau = u(t)$
8. $u(t) * u(t) = \int_{-\infty}^{t} u(\tau)d\tau = r(t)$
9. $r(t) * u(t) = \int_{-\infty}^{t} r(\tau)d\tau = p(t)$
10. $u(t-a) * u(t-b) = r(t-a-b)$
11. $u(t-a) * r(t-b) = p(t-a-b)$

Convolution of two rectangular pulses:

1. Rect. pulses of same duration:

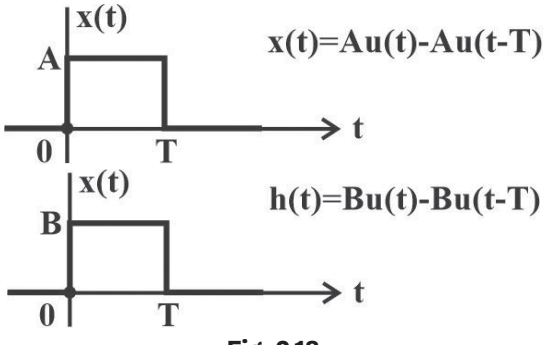


Fig. 2.18

$y(t) = [Au(t) - Au(t-T)]$
$*[Bu(t) - Bu(t-T)]$

$y(t) = Au(t) * Bu(t) - Au(t) *$
$Bu(t-T) - Au(t-T) * Bu(t)$
$+Au(t-T) * Bu(t-T)$

$$y(t) = ABr(t) - ABr(t-T)$$
$$-ABr(t-T) + ABr(t-2T)$$
$$y(t) = ABr(t) - 2ABr(t-T)$$
$$+ABr(t-2T)$$

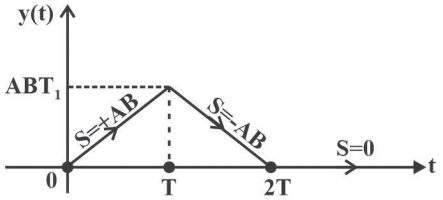


Fig. 2.19

Shortcut:

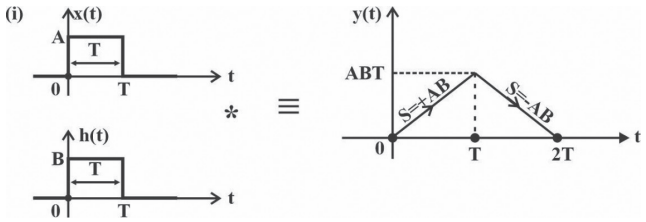


Fig. 2.20

Initial slope: AB = +ve

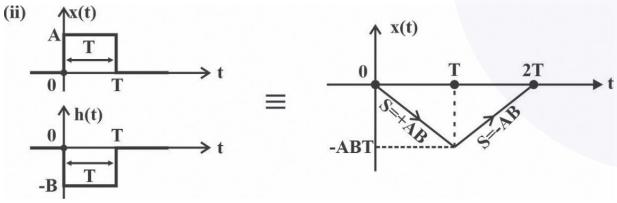


Fig. 2.21

Initial slope: -AB = -ve

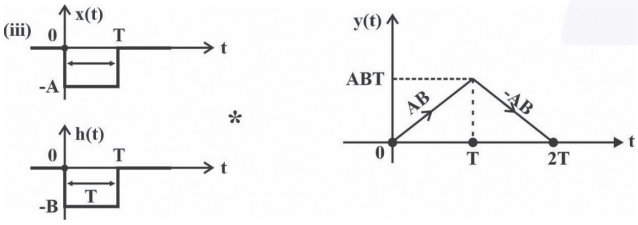


Fig. 2.22

Initial slope $= -A \times -B = AB$

Example:

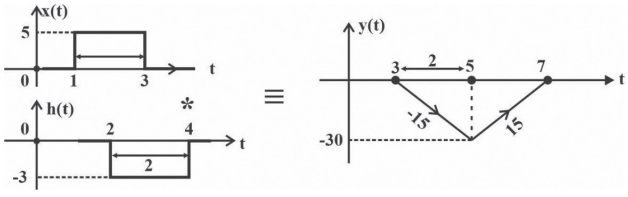


Fig. 2.23

Initial slope = -15

2. Rectangular pulses of different duration:

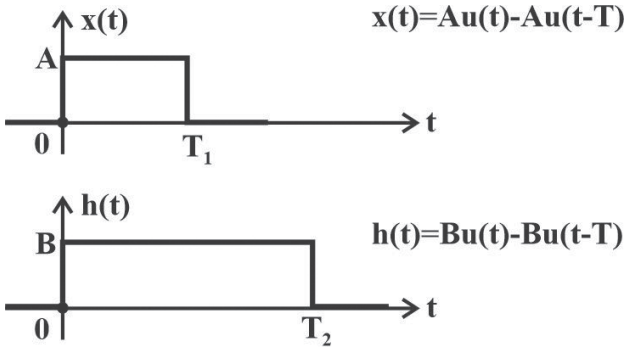


Fig. 2.24

$$y(t) = \left[Au(t) - Au\left(t-T_1\right)\right] * \left[Bu(t) - Bu\left(t-T_2\right)\right]$$
$$y(t) = ABr(t) - ABr\left(t-T_1\right)$$
$$-ABr\left(t-T_2\right) + ABr\left(t-T_1-T_2\right)$$

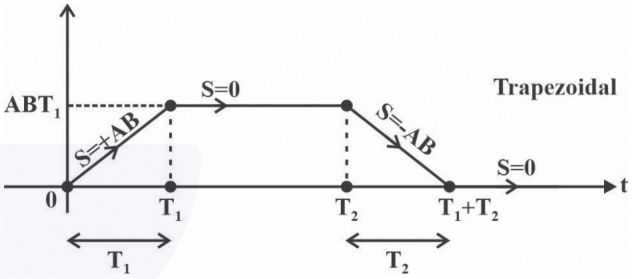


Fig. 2.25

Shortcut:

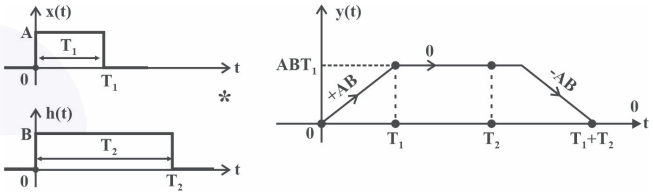


Fig. 2.26

Initial slope = AB = +ve

Example

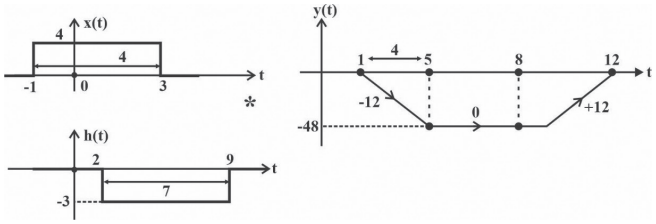


Fig. 2.27

Initial Slope = -12

Convolution of two signals:

Type 1: Convolution of two rectangular pulses.

10. $x(t) = [u(t+2) - u(t-3)]$ & $h(t)$
$= [u(t) - u(t-4)]$

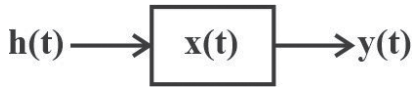


Fig. 2.28

Plot y(t) =?

Solution:

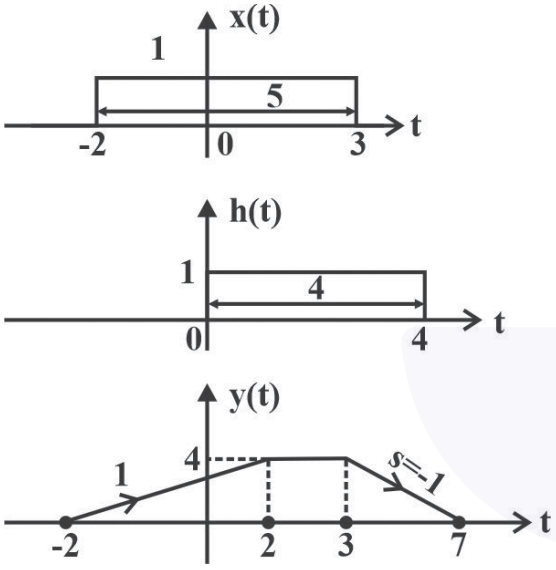


Fig. 2.29

Type 2: One signal is impulse or shifted Impulse

11. $\frac{\sin \pi t}{t} * \delta(t-2) = ?$

Solution: $\frac{\sin \pi t}{t} * \delta(t-2) = \frac{\sin \pi(t-2)}{(t-2)}$

Type 3: If one signal is unit step or shifted unit step:

Step 1→ Identify the range of convolution.

Step 2→ Convert shifted unit step to unit step.

Step 3→ Use standard results.

Example 2.10: $x(t) = e^{-t}u(t) * u(t) = ?$

Solution: Range of convolution: $0 \le t < \infty$

$$g(t) * u(t) = \int_{-\infty}^{t} g(\lambda)d\lambda$$

$$x(t) = e^{-t}u(t) * u(t) = \int_{-\infty}^{t} e^{-\lambda}u(t)d\lambda$$

$$= \int_{-\infty}^{t} e^{-\lambda}d\lambda = \left(1 - e^{-t}\right)$$

$$x(t) = \left(1 - e^{-t}\right) \quad : \quad 0 \le t < \infty$$

$$x(t) = \left(1 - e^{-t}\right)u(t)$$

Type 4:

1. When mathematical expressions are given.
2. $y(t) = x(t) * h(t)$
   Use timeline method:

Timeline method:

Step 1→ Boundary of convolution

Step 2→ $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$
$= \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$

Step 3→ $\left.\begin{matrix} x(\tau) \\ h(t-\tau) \end{matrix}\right]$ → Define mathematically →

Plot on timeline

Step 4→ Vary "t" and calculate y(t) =?

13. $y(t) = e^{-at}u(t) * e^{-bt}u(t) = ?$

Solution:

Step 1→ $y(t) : 0 \le t < \infty$

Step 2→ $y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

Step 3→ $x(t) = e^{-at}u(t)$

$$x(\tau) = e^{-a\tau}u(\tau) = \begin{cases} e^{-a\tau} & 0 \le \tau < \infty \\ 0 & else \end{cases}$$

$$h(t) = e^{-bt}u(t)$$

$$h(t-\tau) = e^{-b(t-\tau)}u(t-\tau) = \begin{cases} e^{-b(t-\tau)} & : \quad -\infty \le \tau \le t \\ 0 & : \quad \text{else} \end{cases}$$

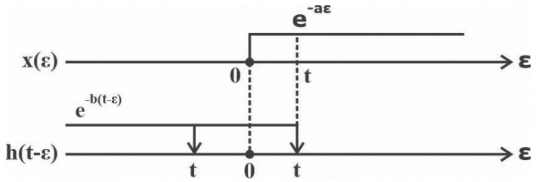


Fig. 2.30

$$t < 0 : \quad \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = 0$$

$$t \ge 0 : \quad \int_{0}^{t} e^{-a\tau}e^{-bt}e^{b\tau}d\tau = e^{-bt}\int_{0}^{t} e^{-(a-b)\tau}d\tau$$

$$= e^{-bt}\frac{\left[e^{(b-a)t} - 1\right]}{(b-a)} = \left\{\frac{e^{-at} - e^{-bt}}{(b-a)}\right\}$$

$$y(t) = \begin{cases} 0 & t < 0 \\ \dfrac{e^{-at} - e^{-bt}}{b-a} & t \geq 0 \end{cases}$$

$$\boxed{y(t) = \left\{\frac{e^{-at} - e^{-bt}}{b-a}\right\} u(t)}$$

Type 5: "Differentiation and integration method"

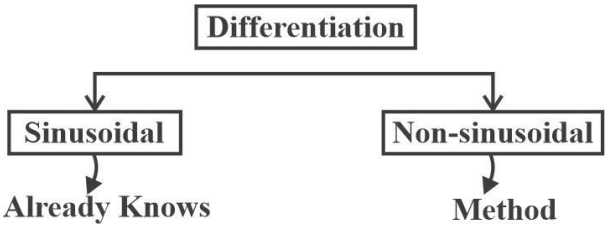


Fig. 2.31

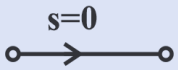
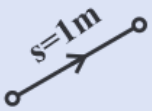
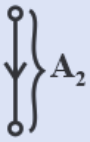
x(t)	Slope	dx(t)/dt
	$s = 0$	→ Part of time axis
	$s = +m$	
	$s = +\infty$	Upward impulse of area A_1
	$s = -\infty$	Downward impulse of area $-A_2$

Table 2.1

Example:

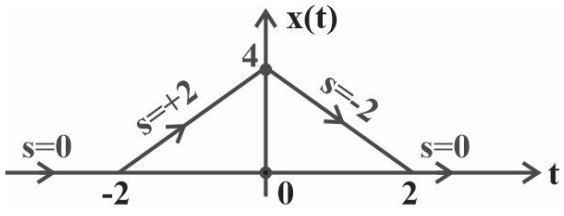


Fig. 2.32

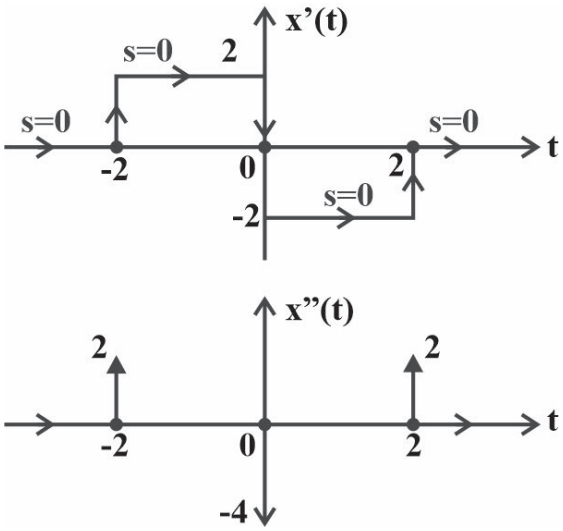


Fig. 2.33

Integration of signals:

$$y(t) = \int_{-\infty}^{t} x(t)dt$$

Case 1: When the signal has step variations only.

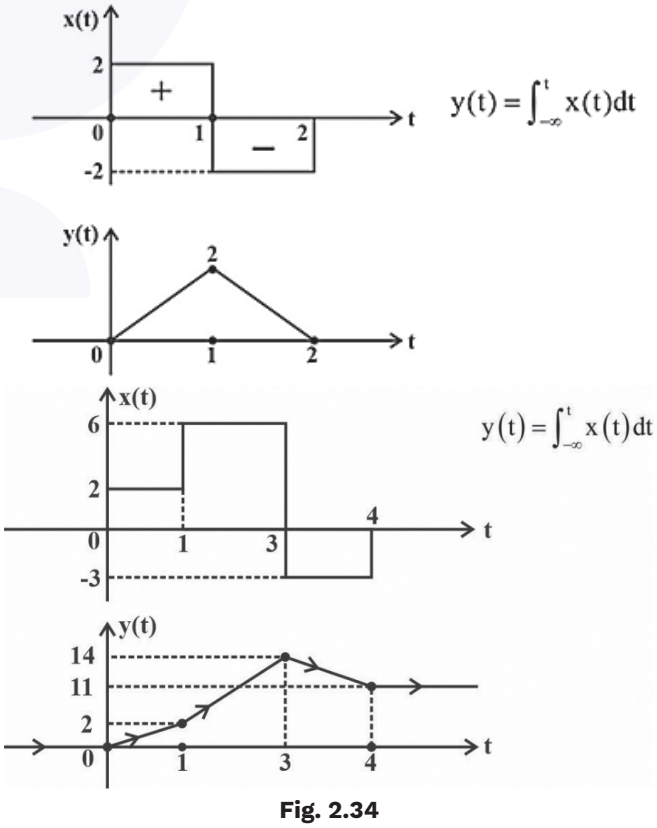


Fig. 2.34

Case 2: If other variations are present along with the step.

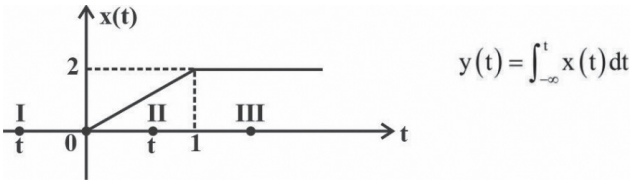


Fig. 2.35

$$x(t)=\begin{cases}0 & : \ t<0\\ 2t & : \ 0\le t<1\\ 2 & : \ t\ge 1\end{cases}$$

$$y(t)=\int_{-\infty}^{t} x(t)dt=\begin{cases}0 & t<0\\ t^2 & 0\le t<1\\ 2t-1 & 1<t<\infty\end{cases}$$

$t<0: \quad y(t)=\int_{-\infty}^{t} x(t)dt=0$

$0\le t<1: \quad y(t)$

$=\int_{-\infty}^{t} x(t)dt=\int_{-\infty}^{0} 0dt+\int_{0}^{t} 2t\ dt=t^2$

$t\ge 1: \quad y(t)$

$=\int_{-\infty}^{t} x(t)dt=\int_{-\infty}^{0} 0dt+\int_{0}^{t} 2t\ dt$

$+\int_{1}^{t} 2dt=1+2t-2=(2t-1)$

Type 5: $y(t)=x(t)*h(t)$ [h(t) has only step variation]

$\frac{dh(t)}{dt}\rightarrow$ Impulse

$z(t)=\frac{dy(t)}{dt}=x(t)*\frac{dh(t)}{dt}\rightarrow$ Convolution operator dissolves.

$y(t)=\int_{-\infty}^{t} z(t)dt$

14.

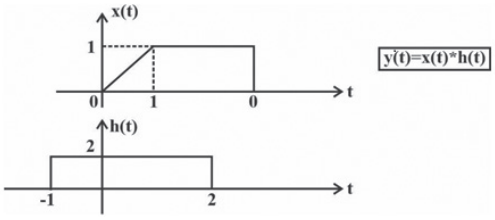


Fig. 2.36

$h(t)=2u(t+1)-2u(t-2)$

$\frac{dh(t)}{dt}=2\delta(t+1)-2\delta(t-2)$

$y(t)=x(t)*h(t)$

$z(t)=\frac{dy(t)}{dt}=x(t)*\frac{dh(t)}{dt}$

$z(t)=2x(t+1)-2)c(t-2)$

$y(t)=\int_{-\infty}^{t} z(t)dt$

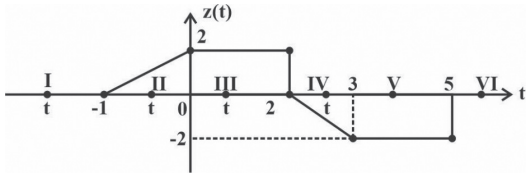


Fig. 2.37

$$Z(t)=\begin{cases}0 & : \ t<-1\\ 2(t+1) & : \ -1<t<0\\ 2 & : \ 0<t<2\\ -2(t-2) & : \ 2<t<3\\ -2 & : \ 3<t<5\\ 0 & : \ t>5\end{cases}$$

$y(t)=\int_{-\infty}^{t} x(t)dt$

$t<-1: \quad \int_{-\infty}^{t} x(t)dt=0$

$-1<t<0: \quad \int_{-\infty}^{t} x(t)dt=\int_{-1}^{t} 2(t+1)dt=(t+1)^2$

$0<t<2: \quad \int_{-\infty}^{t} x(t)dt=\int_{-1}^{0} x(t)dt+\int_{0}^{t} 2dt=2t+1$

$2<t<3: \quad \int_{-\infty}^{t} x(t)dt=$

$\int_{-1}^{0} x(t)dt+\int_{0}^{2} 2dt+$

$\int_{2}^{t} -2(t-2)dt=-t^2+4t+1$

$3<t<5: \quad \int_{-\infty}^{t} x(t)dt=\int_{-1}^{0} x(t)dt+$

$\int_{0}^{2} 2dt+\int_{2}^{3} -2(t-2)dt+$

$\int_{3}^{t} -2dt=10-2t$

$t>5: \quad y(t)=\int_{-\infty}^{t} x(t)dt=0$

$$y(t)=\begin{cases}0 & : \ t<-1\\ (t+1)^2 & : \ -1<t<0\\ (2t+1) & : \ 0<t<1\\ -t^2+4t+1 & : \ 2<t<3\\ (10-2t) & : \ 3<t<5\\ 0 & : \ t>5\end{cases}$$

Type 6:

Graphical method:

S.1→ Range of convolution

S.2→ $y(t)=\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$x(t)\,vs\ t \longrightarrow x(\tau)\,vs\ \tau$

$h(t)\,vs\ t \longrightarrow h(\tau)\,vs\ \tau \xrightarrow{Fold} h(-\tau)\,vs\ \tau$

Right shift by t

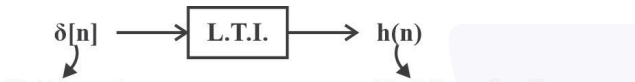
$h(t-\tau)\,vs\ \tau$

$y(t)=x(t)*h(t)$

$y(t)=\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

$y(t)\big|_{t=t_0} = y(t_0)=\int_{-\infty}^{\infty} x(\tau)h(t_0-\tau)d\tau$

Discrete time LTI system:



$\delta(n-1)\longrightarrow h(n-1)$

$x(1)\delta(n-1)\longrightarrow x(1)h(n-1)....(i)$

$x(2)\delta(n-2)\longrightarrow x(2)h(n-2)....(ii)$

$\sum_{k=-\infty}^{\infty} x(k)\delta(n-k)\longrightarrow \sum_{k=-\infty}^{\infty} x(k)h(n-k)$

$x(n)\xrightarrow{LTI\ S/S} \sum_{k=-\infty}^{\infty} x(k)h(n-k)$

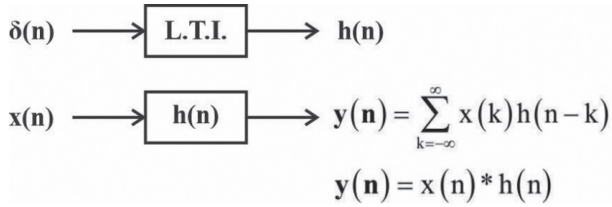


Fig. 2.38

Some standard results:

1. $x(n)*h(n)=\sum_{k=-\infty}^{\infty} x(k)h(n-k)$
$=\sum_{k=-\infty}^{\infty} x(n-k)h(k)$
2. $x(n)*\delta(n)=x(n)$
3. $x(n-n_1)*\delta(n-n_2)=x(n-n_1-n_2)$
4. $\delta(n-n_1)*\delta(n-n_2)=\delta(n-n_1-n_2)$
5. $x(n)*u(n)=\sum_{k=-\infty}^{n} x(k)$
6. $u(n)*u(n)=\sum_{k=-\infty}^{n} u(k)=r(n+1)$
7. $u(n-n_1)*u(n-n_2)=r(n-n_1-n_2+1)$

Type 1: One signal is an impulse or shifted impulse:

15. $y(n)=\left(\frac{1}{2}\right)^n u(n)*\delta(n-3)$ =?

Solution:

$$y(n)=\left(\frac{1}{2}\right)^{n-3}u(n-3)$$

Type 2: One signal is a step or shifted step:

16. $y(n)=\left(\frac{1}{2}\right)^n u(n)*u(n)$ =?

Solution: $y(n): 0\le n<\infty$

$$y(n)=\sum_{k=-\infty}^{n}\left(\frac{1}{2}\right)^k u(k)=\sum_{k=0}^{n}\left(\frac{1}{2}\right)^k$$
$$=1+\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)^2+...+\left(\frac{1}{2}\right)^n$$
$$y(n)=\frac{1\cdot\left[1-(1/2)^{n+1}\right]}{[1-1/2]}\rightarrow y(n)$$
$$=\frac{\left(1-(1/2)^{n+1}\right)}{(1-1/2)}u(n)$$

Type 3: Time line method:

17. $y(n)=\left(\frac{1}{3}\right)^n u(n)*u(3-n)$ =?

Solution:

Step 1→ $y(n): -\infty<n<\infty$

Step 2→ $y(n)=\sum_{k=-\infty}^{\infty} x(k)h(n-k)$

$$x(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$x(k) = \left(\frac{1}{3}\right)^k u(k) = \begin{cases} \left(\frac{1}{3}\right)^k : & k \geq 0 \\ 0 & k \leq -1 \end{cases}$$

$$h(n) = u(3-n)$$

$$h(n-k) = u(3-n+k) = \begin{cases} 1 & n-3 \leq k < \infty \\ 0 & \text{elese} \end{cases}$$

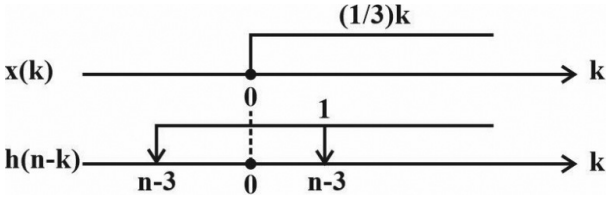


Fig. 2.39

$$n-3<0: \quad n-3 \leq -1 \quad \sum_{K=0}^{\infty}\left(\frac{1}{3}\right)^K = \frac{3}{2}$$

$$n-3 \geq 0: \quad n-3 \geq 0 \quad \sum_{k=n-3}^{\infty}\left(\frac{1}{3}\right)^k$$

$$= \left(\frac{1}{3}\right)^{n-3} + \left(\frac{1}{3}\right)^{n-2} + \left(\frac{1}{3}\right)^{n-1} + \left(\frac{1}{3}\right)^{n} + \left(\frac{1}{3}\right)^{n+1}$$

$$= \left(\frac{1}{3}\right)^n \{27+9+3+1+1/3+1/9+\cdots]$$

$$= (1/3)\frac{n}{1-1/3}$$

$$y(n) = \begin{cases} \frac{3}{2} & n \leq 2 \\ \frac{1}{2}3^{4-n} & n \geq 3 \end{cases}$$

$$y(n) = \frac{3}{2}u[-n+2] + \frac{1}{2}3^{4-n}u[n-3]$$

Type 4: Convolution of a finite duration discrete-time signal.

Method 1: $y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} x(n-k)h(k)$

$$= \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Method 2: Tabular method

4. x(n)={5,-1,$\underset{\uparrow}{2}$,4}

h(n)={1,$\underset{\uparrow}{2}$,3}

Or

$$x(n) = 5\delta(n+2) - \delta(n+1) + 2\delta(n) + 4\delta(n-1)$$

$$h(n) = \delta(n+1) + 2\delta(n) + 3\delta(n-1)$$

Or

$$x(n) = 5\ u(n+2) - 6\ u(n+1) + 3\ u(n)$$
$$+2\ u(n-1) - 4\ u(n-2)$$

$$h(n) = u(n+1) + u(n) + u(n-1) - 3u(n-2)$$

	5	-1	2	4
1	5	-1	2	4
2	10	-2	4	8
3	15	-3	6	12

Fig. 2.40

$$y(n) = \left\{5,\ 9,\ 15,\ \underset{\uparrow}{5},\ 14,\ 12\right\}$$

$$\sum x(n) = 10$$

$$\sum h(n) = 6$$

$$\sum y(n) = \sum x(n) * \sum h(n) = 60$$

Properties:

N_1: Largest no in x(n)

N_2: Largest no in h(n)

1. $x(n) = l$

   $h(n) = m$

   $y(n) = l + m - 1$

2. $\sum y(n) = \sum x(n) \times \sum h(n)$

3. $\{\sum y(n)\}_{max} = \{\sum x(n)\}_{max} \times \{\sum h(n)\}_{max}$

   $= lN_1 \times mN_2$

Example 2.18: $x(n) = \left\{\underset{-2}{1},\ \underset{-1}{-1},\ \underset{\uparrow 0}{3},\ \underset{1}{0},\ \underset{2}{2}\right\}$ &

$$h(n) = \left\{\underset{-2}{-2},\ \underset{-1}{0},\ \underset{\uparrow 0}{1},\ \underset{1}{2}\right\}$$

Calculate x(n)*h(n)

Solution:

$$x(n)=\left\{\underset{-2}{1},\ \underset{-1}{-1},\ \underset{\uparrow 0}{3},\ \underset{1}{0},\ \underset{2}{2}\right\}: l=5$$

$$h(n)=\left\{\underset{-2}{-2},\ \underset{-1}{0},\ \underset{\uparrow 0}{1},\ \underset{1}{2}\right\}: \mathrm{m}=4$$

	1	0	3	0	2
-2	-2	0	-6	0	-4
0	0	0	0	0	0
1	1	0	3	0	2
2	2	0	6	0	4

Fig. 2.41

$$y(n)\left\{\underset{-4}{-2},\ 0,\ -5,\ 2,\ \underset{\uparrow}{-1},\ \underset{1}{6},\ \underset{2}{2},\ \underset{3}{4}\right\}$$

$$l+m-1=8$$

Important category:

$$x(n)=x_1(n)u(n)$$

$$h(n)=x_2(n)u(n)$$

$$y(n)=x(n)*h(n)$$

$$y(n)|_{n=n_0}=?$$

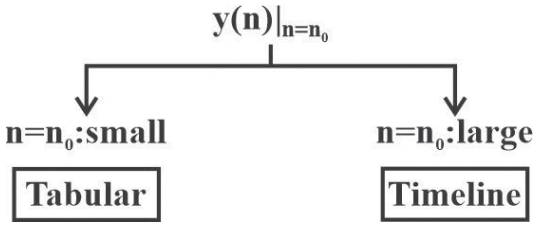


Fig. 2.42

19. $h(n)=\left(\frac{1}{2}\right)^n u(n)$

$y(n)=x(n)*h(n)$

$y(0)=1$

$y(1)=\frac{1}{2}$

Where x(n) is causal signal: calculate x (1) =?

Solution:

x(n) is causal: $y(n)=x(n)*h(n)$

$$\{y(0),y(1)\ldots..\}=\{x(0),x(1)\ldots\}*\{h(0),h(1)\cdots\}$$

$$y(n)=\left\{\underset{\uparrow}{1},\ \frac{1}{2},\ y(2)\right\}$$

$$x(n)=\left\{x(0),\ x(1),\ x(2)\right\}$$

$$h(n)=\left\{\underset{\uparrow}{1},\ \frac{1}{2},\ \frac{1}{2}\right\}$$

	x(0)	x(1)	x(2)
1	x(0)	x(1)	x(2)
1/2	0.5x(0)	0.5x(1)	0.5x(2)
1/4	0.25x(0)	0.25x(1)	0.25x(2)

Fig. 2.43

→ in table signal names missed.

$$y(0)=x(0)=1$$

$$y(1)=x(1)+0.5x(0)$$

$$\frac{1}{2}=x(1)+\frac{1}{2}$$

$$x(1)=0$$

Properties of LTI system:

1. Static nature of LTI system:

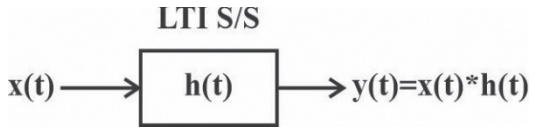


Fig. 2.44

For LTI system to be static y(t) = Ax(t).

It is possible only when $h(t)=A\delta(t)$

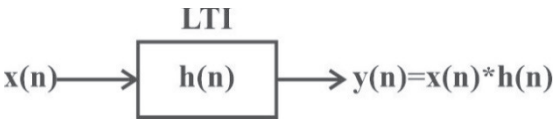


Fig. 2.45

For LTI system to be static y(n) = Ax(n).

It is possible only when $h(n)=A\delta(n)$

Note: "For an LTI system to be static system, the unit impulse response h(t)/ h(n) must be an impulse signal present at origin"

Causality of an LTI system:

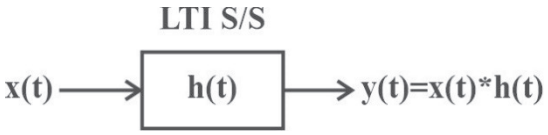


Fig. 2.46

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

Case1:

$t = 4, \tau = 2$ $\quad y(4) = \int_{-\infty}^{\infty} x(2)h(t-\tau)d\tau$

Output depends on the past value of input.

Represents a causal system

Case 2:

$t = 4, \tau = 4$ $\quad y(4) = \int_{-\infty}^{\infty} x(4)h(t-\tau)d\tau$

Output depends on the present value of input.

Represents a causal s/s

Case 3:

$t = 2, \tau = 4$ $\quad y(2) = \int_{-\infty}^{\infty} x(4)h(t-\tau)d\tau$

Output depends on the future value of input.

Represents a non-causal s/s

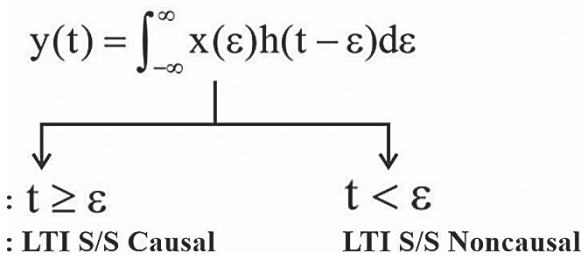


Fig. 2.47

Requirement: "A LTI s/s is to be designed such that it is always causal system."

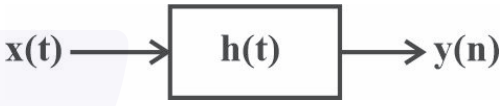


Fig. 2.48

Chapter Summary

1. $s(t) = \int_{-\infty}^{t} h(t)dt$ $\quad h(t) = \frac{ds(t)}{dt}$

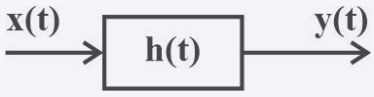


Fig. 2.49

$$y(t) = x(t) * h(t)$$

$$y(t) = x(t) * \frac{ds(t)}{dt}$$

2. $s(n) = \sum_{k=-\infty}^{n} n(k), \quad h(n) = s(n) - s(n-1)$

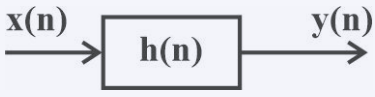


Fig. 2.50

$$y(n) = x(n) * h(n)$$

$$y(n) = x(n) * [s(n) - s(n-1)]$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

→ Non-causal s/s for $t < \tau$:

→ If $h(t-\tau) = 0$ For $t < \tau$ → Then the above system always remains causal.

For LTI system to be causal system:

$h(t-\tau) = 0 \quad t < \tau$

$h(t-\tau) = 0 \quad t - \tau < 0$

$h(p) = 0 \qquad p < 0$

$h(t) = 0$ For $t < 0$

Note: "For an LTI system to be causal system, the unit impulse response h(t)/ h(n) must be a casual signal"

Note: For an LTI system, the comment on causality can be done with impulse response h(t) itself, whereas for any general system, it can be done with the basic definition of the causal system, i.e., at any instant, if the output depends on present and past value of the input, then the system is causal system else it is a non-causal system.

LTI systems in parallel:

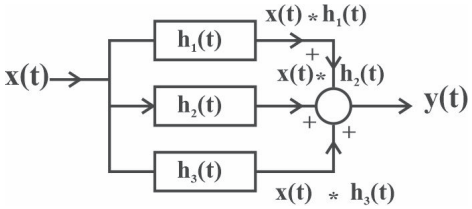
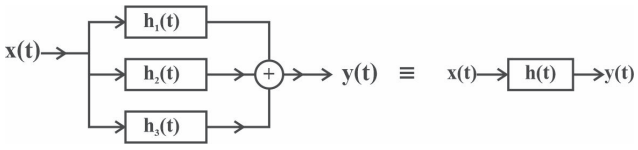


Fig. 2.51

$$y(t) = x(t) * h_1(t) + x(t) * h_2(t) + x(t) * h_3(t)$$

$$y(t) = x(t) * \{h_1(t) + h_2(t) + h_3(t)\}$$



$$h(t) = h_1(t) + h_2(t) + h_3(t)$$

LTI Systems in cascade:

$x(t) * h_1(t)$

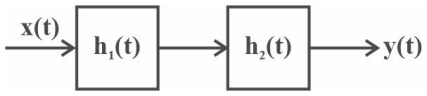


Fig. 2.52

$x(t) * (h_1(x) * h_2(x))$

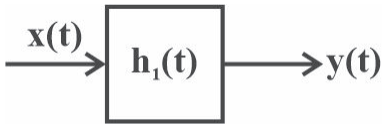


Fig. 2.53

$h(t) * (h_1(x) * h_2(x))$

Unit step response of an LTI system:

1.

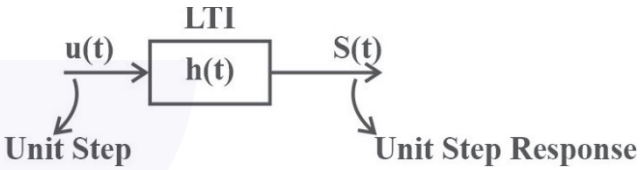


Fig. 2.54

$$s(t) = u(t) * h(t)$$

$$s(t) = \int_{-\infty}^{t} h(t)dt \rightarrow h(t) = \frac{ds(t)}{dt}$$

2.

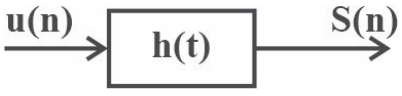


Fig. 2.55

$$s(n) = u(n) * h(n)$$

$$\rightarrow s(n) = \sum_{k=-\infty}^{n} h(k) = h(n) + \sum_{k=-\infty}^{n-1} h(k) \qquad \ldots \text{(i)}$$

$$\rightarrow s(n) = \sum_{k=-\infty}^{n} h(k) \qquad \ldots \text{(ii)}$$

$$h(n) = S(n) - S(n-1)$$

Practice Questions

1. An excitation is applied to a system at $t = T$ and its response is zero for $-\infty < t < T$. Such a system is a

(A) non-causal system

(B) stable system

(C) causal system

(D) unstable system

2. A continuous-time system is governed by the equation:

$3y^3(t) + 2y^2(t) + y(t) = x^2(t) + x(t)$

{$y(t)$ and $x(t)$ respectively are output and input}. The system is

(A) linear and dynamic

(B) linear and non-dynamic

(C) non-linear and dynamic

(D) non-linear and non-dynamic

3. The input and output of a continuous time system are respectively denoted by $x(t)$ and y (t). Which of the following descriptions corresponds to a causal system?

(A) $y(t) = x(t-2) + x(t+4)$

(B) $y(t) = (t-4)x(t+1)$

(C) $y(t) = (t+4)x(t-1)$

(D) $y(t) = (t+5)x(t+5)$

4. If the response of a system to an input does not depend on the future values of the input, then which one of the following is true for the system?

(A) It is aperiodic (B) It is causal

(C) It is anticipatory (D) It is discrete

5. Which one of the following systems is a causal system? [$y(t)$ is output and $u(t)$ is an input step function]

(A) $y(t) = \sin(u(t+3))$

(B) $y(t) = 5u(t) + 3u(t-1)$

(C) $y(t) = 5u(t) + 3u(t+1)$

(D) $y(t) = \sin(u(t-3)) + \sin(u(t+3))$

Key and Explanations

1. **(A)** For a s/m to be casual,

$T = 0$

Then, $x(t) = 0, \quad t < 0$

$\neq 0, \quad t > 0$

But there T it is not mentioned that $T = 0$,

Hence the most appropriate option is 'A'.

2. **(D)** For the s/m to be linear, the power of $x(t)$ & $y(t)$ should be 1. Since, the power of $x(t)$ & $y(t)$ is more than '1'it is non-linear. Also, here present output depends on present input; hence the s/m is non-dynamic Viz Static.

3. **(C)** For the s/m to be casual present output should depend on present input or past input but not on the future values of input.

Hence, option 'C' is correct.

4. **(B)** For the s/m to be casual present output should depend on present input or past input, but not on the future values of input.

Hence, option 'B' is correct.

5. **(C)** For the s/m to be casual present output should depend on present input or past input, but not on the future values of input.

Hence, option 'C'is correct.

Answer Key

1.	**A**	**2.**	**D**	**3.**	**C**	**4.**	**B**	**5.**	**C**